Sets

In [ ]:
```
1.Unordered & Unidexed collection of items.
2.Set elements are unique.Duplicate elements are not allowed.
3.Set elements are immutable(cannot be changed).
4.Set itsekf is mutable.We can add or remove items from it.
```

Set Creation

In [256…
```
myset={1,2,3,4,5}
myset
```

Out[256…
```
{1, 2, 3, 4, 5}
```

In [257…
```
len(myset)
```

Out[257…
```
5
```

In [258…
```
my_set={1,1,2,2,3,4,5,5}
my_set
```

Out[258…
```
{1, 2, 3, 4, 5}
```

In [259…
```
myset1={1.79,2.08,3.99,4.56,5.45}
myset1
```

Out[259…
```
{1.79, 2.08, 3.99, 4.56, 5.45}
```

In [260…
```
myset2={'Asif','John','Tyrion'}
myset2
```

Out[260…
```
{'Asif', 'John', 'Tyrion'}
```

In [261…
```
myset3={10,20,"Aisha",(11,22,32)}
myset3
```

Out[261…
```
{(11, 22, 32), 10, 20, 'Aisha'}
```

In [262…
```
myset3={10,20,"Aisha",[11,22,32]}
myset3
```

```
---------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[262], line 1
----> 1 myset3={10,20,"Aisha",[11,22,32]}
      2 myset3

TypeError: unhashable type: 'list'
```

In [263…

```
myset4=set()
print(type(myset4))
```

```
<class 'set'>
```

my_set1=set(('one','two','three','four')) my_set1

Loop through a set

In [264…

```
myset={'one','two','three','four','five','six','seven','eight'}
for i in myset:
    print(i)
```

```
one
four
two
seven
six
three
five
eight
```

for i in enumerate(myset): print(i)

Set Membership

In [265…

```
myset
```

Out[265…

```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [266…

```
'one' in myset
```

Out[266…

```
True
```

In [267…

```
'ten' in myset
```

Out[267…

```
False
```

In [268…

```
if 'three' in myset:
    print('Three is present in the set')
else:
    print('Three is not present in the set')
```

```
Three is present in the set
```

In [269…

```
if'eleven' in myset:
    print('eleven is present in the set')
else:
    print('eleven is not present in the set')
```

```
eleven is not present in the set
```

Add & Remove Items

In [270…
```
myset
```

Out[270…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [271…
```
myset.add('NINE')
myset
```

Out[271…
```
{'NINE', 'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [272…
```
myset.update(['TEN','ELEVEN','TWELVE'])
myset
```

Out[272…
```
{'ELEVEN',
 'NINE',
 'TEN',
 'TWELVE',
 'eight',
 'five',
 'four',
 'one',
 'seven',
 'six',
 'three',
 'two'}
```

In [273…
```
myset.remove('NINE')
myset
```

Out[273…
```
{'ELEVEN',
 'TEN',
 'TWELVE',
 'eight',
 'five',
 'four',
 'one',
 'seven',
 'six',
 'three',
 'two'}
```

In [274…
```
myset.discard('TEN')
myset
```

Out[274…
```
{'ELEVEN',
 'TWELVE',
 'eight',
 'five',
 'four',
 'one',
 'seven',
 'six',
 'three',
 'two'}
```

In [275…
```
myset.clear()
myset
```

Out[275…
```
set()
```

In [276…
```
del myset
myset
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[276], line 2
      1 del myset
----> 2 myset

NameError: name 'myset' is not defined
```

Copy Set

In [277…
```
myset={'one','two','three','four','five','six','seven','eight'}
myset
```

Out[277…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [278…
```
myset1=myset
myset1
```

Out[278…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [279…
```
id(myset),id(myset1)
```

Out[279…
```
(2493115539840, 2493115539840)
```

In [280…
```
my_set=myset.copy()
my_set
```

Out[280…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

In [281…
```
id(my_set)
```

Out[281…
```
2493115540736
```

In [282…
```
myset.add('nine')
myset
```

Out[282…
```
{'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}
```

In [283…
```
myset1
```

Out[283…
```
{'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}
```

In [284…
```
my_set
```

Out[284…
```
{'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

Set Operation

Union

In [285…
```
A={1,2,3,4,5}
B={4,5,6,7,8}
C={8,9,10}
```

In [286…
```
A | B
```

Out[286…
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [287…
```
A.union(B)
```

Out[287…
```
{1, 2, 3, 4, 5, 6, 7, 8}
```

In [288…
```
A.union(B,C)
```

Out[288…
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

In [289…
```
A.update(B,C)
A
```

Out[289…
```
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

In [ ]:

Intersection

In [290…
```
A={1,2,3,4,5}
B={4,5,6,7,8}
```

In [291…
```
A&B
```

Out[291… {4, 5}

In [292…
```
A.intersection(B)  Intersection A and B
```

```
  Cell In[292], line 1
    A.intersection(B)  Intersection A and B
                       ^
SyntaxError: invalid syntax
```

In [293…
```
A.intersection_update(B)
A
```

Out[293… {4, 5}

In [ ]:

Difference

In [294…
```
A={1,2,3,4,5}
B={4,5,6,7,8}
```

In [295…
```
A-B  #set of elements that are only in A but not in B
```

Out[295… {1, 2, 3}

In [296…
```
A.difference(B)
```

Out[296… {1, 2, 3}

In [297…
```
B-A
```

Out[297… {6, 7, 8}

In [298…
```
B.difference(A)
```

Out[298… {6, 7, 8}

In [299…
```
B.difference_update(A)
B
```

Out[299… {6, 7, 8}

In [ ]:

Symmetric Difference

In [300…
```
A={1,2,3,4,5}
B={4,5,6,7,8}
```

In [301…
```
A ^ B
```
Out[301…
```
{1, 2, 3, 6, 7, 8}
```
In [302…
```
A.symmetric_difference(B)
```
Out[302…
```
{1, 2, 3, 6, 7, 8}
```
In [303…
```
A.symmetric_difference_update(B)
A
```
Out[303…
```
{1, 2, 3, 6, 7, 8}
```
In [ ]:
```

```

Subset,Superset&Disjoint

In [304…
```
A={1,2,3,4,5,6,7,8,9}
B={3,4,5,6,7,8}
C={10,20,30,40}
```
In [305…
```
B.issubset(A)
```
Out[305…
```
True
```
In [306…
```
A.issuperset(B)
```
Out[306…
```
True
```
In [307…
```
C.isdisjoint(A)
```
Out[307…
```
True
```
In [308…
```
B.isdisjoint(A)
```
Out[308…
```
False
```

Other Builtin functions

In [309…
```
A
```
Out[309…
```
{1, 2, 3, 4, 5, 6, 7, 8, 9}
```
In [310…
```
sum(A)
```
Out[310…
```
45
```
In [311…
```
max(A)
```

Out[311… 9

In [312…
```
min(A)
```

Out[312… 1

In [313…
```
len(A)
```

Out[313… 9

In [314…
```
list(enumerate(A))
```

Out[314… [(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 9)]

In [315…
```
D=sorted(A,reverse=True)
D
```

Out[315… [9, 8, 7, 6, 5, 4, 3, 2, 1]

In [316…
```
sorted(D)
```

Out[316… [1, 2, 3, 4, 5, 6, 7, 8, 9]

Dictionary

In [317…
```
-Dictionary is a mutable data type in python.
-A python dictionary is a collection of key and value pairs separated by a colon(:)
-Keys must be unique in a dictionary,duplicate values are allowed.
```

```
  Cell In[317], line 1
    -Dictionary is a mutable data type in python.
                     ^
SyntaxError: invalid syntax
```

In [318…
```
mydict={'A':'ONE','B':'TWO','C':'THREE','D':'FOUR'}
mydict
```

Out[318… {'A': 'ONE', 'B': 'TWO', 'C': 'THREE', 'D': 'FOUR'}

In [319…
```
mydict=dict()
mydict
```

Out[319… {}

In [320…
```
mydict={}
mydict
```

Out[320… {}

In [321…
```
mydict={1:'ONE',2:'TWO',3:'THREE'}
mydict
```

Out[321…
```
{1: 'ONE', 2: 'TWO', 3: 'THREE'}
```

In [322…
```
mydict=dict({1:'ONE',2:'TWO',3:'THREE'})
mydict
```

Out[322…
```
{1: 'ONE', 2: 'TWO', 3: 'THREE'}
```

In [323…
```
mydict={'A':'ONE','B':'TWO','C':'THREE'}
mydict
```

Out[323…
```
{'A': 'ONE', 'B': 'TWO', 'C': 'THREE'}
```

In [324…
```
mydict={1:'ONE','A':'TWO',3:'THREE'}
mydict
```

Out[324…
```
{1: 'ONE', 'A': 'TWO', 3: 'THREE'}
```

In [325…
```
mydict.keys()
```

Out[325…
```
dict_keys([1, 'A', 3])
```

In [326…
```
mydict.values()
```

Out[326…
```
dict_values(['ONE', 'TWO', 'THREE'])
```

In [327…
```
mydict.items()
```

Out[327…
```
dict_items([(1, 'ONE'), ('A', 'TWO'), (3, 'THREE')])
```

In [328…
```
mydict={1:'ONE',2:'TWO','A':['aisha','john','Maria'],'B':('Bat','Cat')}
mydict
```

Out[328…
```
{1: 'ONE', 2: 'TWO', 'A': ['aisha', 'john', 'Maria'], 'B': ('Bat', 'Cat')}
```

In [329…
```
mydict={1:'ONE',2:'TWO','A':{'Name':'asif','Age':20},'B':('Bat','Cat','hat')}
mydict
```

Out[329…
```
{1: 'ONE',
 2: 'TWO',
 'A': {'Name': 'asif', 'Age': 20},
 'B': ('Bat', 'Cat', 'hat')}
```

In [330…
```
keys={'a','b','c','d'}
value=10
mydict3=dict.fromkeys(keys,value)
mydict3
```

Out[330…
```
{'a': 10, 'b': 10, 'c': 10, 'd': 10}
```

In [331…
```
keys={'a','b','c','d'}
value=[10,20,30]
```

```
mydict3=dict.fromkeys(keys,value)
mydict3
```

Out[331…
```
{'a': [10, 20, 30], 'b': [10, 20, 30], 'c': [10, 20, 30], 'd': [10, 20, 30]}
```

In [332…
```
value.append(40)
mydict3
```

Out[332…
```
{'a': [10, 20, 30, 40],
 'b': [10, 20, 30, 40],
 'c': [10, 20, 30, 40],
 'd': [10, 20, 30, 40]}
```

Accessing items

In [333…
```
mydict={1:'ONE',2:'TWO',3:'THREE',4:'FOUR'}
mydict
```

Out[333…
```
{1: 'ONE', 2: 'TWO', 3: 'THREE', 4: 'FOUR'}
```

In [334…
```
mydict[1]
```

Out[334…
```
'ONE'
```

In [335…
```
mydict.get(1)
```

Out[335…
```
'ONE'
```

In [336…
```
mydict1={'Name':'Aisha','ID':5114,'DOB':2005,'JOB':'Analyst'}
mydict1
```

Out[336…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005, 'JOB': 'Analyst'}
```

In [337…
```
mydict1['Name']  #Access item using key
```

Out[337…
```
'Aisha'
```

In [338…
```
mydict1.get('JOB')   #Access item using get() method
```

Out[338…
```
'Analyst'
```

Add,Remove & Change Items

In [339…
```
mydict1={'Name':'Aisha','ID':5114,'DOB':2005,'Address':'Odisha'}
mydict1
```

Out[339…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005, 'Address': 'Odisha'}
```

In [340…
```
mydict1['DOB']=2005
mydict1['Address']='Odisha'
```

```
mydict1
```

Out[340…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005, 'Address': 'Odisha'}
```

In [341…
```
dict1={'DOB':2005}
mydict1.update(dict1)
mydict1
```

Out[341…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005, 'Address': 'Odisha'}
```

In [342…
```
mydict1['JOB']='Analyst'
mydict1
```

Out[342…
```
{'Name': 'Aisha',
 'ID': 5114,
 'DOB': 2005,
 'Address': 'Odisha',
 'JOB': 'Analyst'}
```

In [343…
```
mydict1.pop('JOB')
mydict1
```

Out[343…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005, 'Address': 'Odisha'}
```

In [344…
```
mydict1.popitem()
mydict1
```

Out[344…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005}
```

In [345…
```
mydict1
```

Out[345…
```
{'Name': 'Aisha', 'ID': 5114, 'DOB': 2005}
```

In [346…
```
del[mydict1['ID']]
mydict1
```

Out[346…
```
{'Name': 'Aisha', 'DOB': 2005}
```

In [347…
```
mydict1.clear()
mydict1
```

Out[347…
```
{}
```

In [348…
```
del mydict1
mydict1
```

```
---------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[348], line 2
      1 del mydict1
----> 2 mydict1

NameError: name 'mydict1' is not defined
```

Copy Dictionary

In [349…
```
mydict={'Name':'Aisha','ID':12345,'DOB':2005,'Address':'Odisha'}
mydict
```

Out[349…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [350…
```
mydict1=mydict
mydict1
```

Out[350…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [351…
```
id(mydict),id(mydict1)
```

Out[351…
```
(2493113907584, 2493113907584)
```

In [352…
```
mydict2=mydict.copy()
mydict2
```

Out[352…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [356…
```
id(mydict2)
```

Out[356…
```
2493113085696
```

In [357…
```
mydict['Address']='Odisha'
mydict
```

Out[357…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [358…
```
mydict1
```

Out[358…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [359…
```
mydict2
```

Out[359…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

Loop Through a Dictionary

In [360…
```
mydict1={'Name':'Aisha','ID':12345,'DOB':2005,'Address':'Odisha'}
mydict1
```

Out[360…
```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [361…
```
for i in mydict:
    print(i,':',mydict[i])
```

```
Name : Aisha
ID : 12345
DOB : 2005
Address : Odisha
```

In [362…

```
for i in mydict1:
    print(mydict[i])
```

```
Aisha
12345
2005
Odisha
```

Dictionary Membership

In [363…

```
mydict1={'Name':'Aisha','ID':12345,'DOB':2005,'Address':'Odisha'}
mydict1
```

Out[363…

```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [364…

```
'Name' in mydict1
```

Out[364…

```
True
```

In [365…

```
'Aisha' in mydict1
```

Out[365…

```
False
```

In [368…

```
'ID' in mydict1
```

Out[368…

```
True
```

In [367…

```
'Address' in mydict1
```

Out[367…

```
True
```

All/Any

In [ ]:

```
The all() method returns:
-True-if all key of the dictionary are true
-False-if any key of the dictionary is false
The any() functions returns True if any key of the dictionary is True.if not,any()r
```

In [369…

```
mydict1={'Name':'Aisha','ID':12345,'DOB':2005,'Address':'Odisha'}
mydict1
```

Out[369…

```
{'Name': 'Aisha', 'ID': 12345, 'DOB': 2005, 'Address': 'Odisha'}
```

In [370…

```
all(mydict1)
```

Out[370… True

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: